CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 51 (2015) 206–210

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

A thermodynamic assessment of the Nd–C system

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ARTICLE INFO

Article history:
Received 10 June 2015
Received in revised form
1 October 2015
Accepted 1 October 2015
Available online 11 November 2015

Keywords:
Nd–C system
Phase diagram
Thermodynamic modeling

ABSTRACT

The Nd–C binary system has been assessed based on the experimental investigation, thermodynamic calculations and the literature survey. The intermediate compounds Nd_2C_3 and βNdC_2 (high-temperature form of NdC_2 phase) were considered to be the non-stoichiometric phases whereas the αNdC_2 (low-temperature form of NdC_2 phase) was treated as the stoichiometric phase. Contents of carbon in the Nd–C_{sat} samples in Nd–Nd_2C_3 range were measured in the present study. The phase diagram of Nd–C system and enthalpies of formation of Nd_2C_3 and NdC_2 carbides have been modeled using the FactSage software.

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1. Introduction

The increasing popularities of electronic consumer goods, hybrid and electric cars, and wind turbines lead to an unprecedented increase in the demand of rare earth elements (REEs). REEs are now considered as the most critical in EU for the highest supply risk. Because there exist only few exploitable natural resources of rare earths in Europe, the EU will mainly have to rely on recycling of REEs from pre- and post-consumer scraps and especially the End-of-Life (EoL) products.

The phase relations and thermodynamic properties of the Nd–Fe–B–C system, particular the carbon-saturated alloys, are the essential properties for the development and optimization of recovery of REEs in magnet scraps by high temperature processes. The Nd–C binary system is therefore one of the key subsystems. However, this binary system has not yet been well constructed in the literature. In our recent work, carbon contents of the carbon-saturated Nd–C_{sat} samples in the Nd–Nd_2C_3 range have been measured in the laboratory. Using the available equilibrium data for the Nd–C system, we establish the Nd–C phase diagram by the CALPHAD method.

2. Review of experimental data

The phase diagram of Nd–C system is not available in the literatures. Because of the similarity for the chemical and physical properties of the light rare earth elements (La, Ce, Pr and Nd), the experimental determined La–C, Ce–C and Pr–C phase diagrams can be referred as the prototype of the Nd–C system.

According to assessment given by Gschneidner and Calderwood [1], there are two stable carbides existing in the Nd–C system: Nd_2C_3 and NdC_2. Although El Makrini et al. [2] reported the existence of NdC_6 compound in the system, it has not been taken into consideration in the present work for the unclear structure of this compound.

Spedding et al. [3] investigated the phase relation of the Nd–C system between the carbon content at 0–50% by X-ray diffraction. Only the sesquicarbide, Nd_2C_3, and the hexagonal Nd appeared in this composition range. The lattice parameters of Nd sesquicarbide and dicarbide were also given by Spedding et al. [3] It was also observed that Nd_2C_3 exhibits a range of solid solubility, listing lattice parameters for the metal-rich and the C-rich phase [1,3]. The βNdC_2 also exhibits certain homogeneities while αNdC_2 exhibits negligible homogeneity based on the analysis of the lattice parameters. The crystal structure data about all the phases in the Nd–C system are listed in Table 1.

Based on the literature data [1], Gschneidner and Calderwood assessed the cubic to tetragonal transformation temperature of $\alpha NdC_2 \rightarrow \beta NdC_2$, which was 1150 ± 20 °C. Thermochemical properties of the Nd carbides were reported by several authors. The enthalpy of formation of Nd_2C_3 at 298 K was measured by Anderson and Bagshaw [6]. Enthalpy of formation of NdC_2 was reported by Anderson and Bagshaw [7], Faircloth et al. [8], Vidhya et al. [9] and Meschel et al. [10]. Table 2 summarized the data available in the literature.

Based on the assessed prototype of La–C [11,12], Ce–C [12,13] and Pr–C [12,14] systems, it is confirmed that two eutectic

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http://dx.doi.org/10.1016/j.calphad.2015.10.001
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Table 1
Crystal structure data for the solid phases in Nd–C system.

Phase	Pearson symbol	Space group	Prototype	Lattice parameter (Å)			Reference
				a	*b*	*c*	
αNd (DHCP)	hP4	$P6_3/mmc$	αLa	3.66		11.80	[4]
βNd (BCC)	cI2	$Im\bar{3}m$	W	4.13			[4]
Nd_2C_3	cI40	$I\bar{4}3d$	Pu_2C_3	8.52(a)/8.55(b)			[3]
αNdC_2	tI6	$I4/mmm$	CaC_2	3.82		6.39	[3]
βNdC_2	?						[5]
C(Graphite)	hP4	$P6_3/mmc$	C-graphite	2.52		4.12	

Table 2
The enthalpies of formation of the compounds in the Nd–C system.

Compound	ΔH_f (298 K), kJ/mol	Method	Reference
Nd_2C_3	-34.09 ± 3	EMF	[6]
NdC_2	-29.33 ± 3	EMF	[7]
	-17.4 ± 3.3	Knudsen effusion	[8]
	-29.2 ± 1.4	Calorimetry	[9]
	-30.3 ± 2.0	Calorimetry	[10]

reactions occur in the Nd–C system, i.e., $L \rightarrow \beta NdC_2 + C_{gr}$. and $L \rightarrow Nd + Nd_2C_3$. Paderno et al. [15] reported the melting point of Nd_2C_3 to be 1620 ± 30 °C. Faircloth et al. [8] reported the congruent melting temperature of '$NdC_{2.29}$' carbide at 2290 ± 10 °C. Gschneidner and Calderwood [1] assessed temperatures for NdC_2–C eutectic at 2260 ± 30 °C. The melting point of NdC_2 would be higher than 2260 °C. Like other light rare earth carbon binary systems [11–14], peritectic reaction ($L + \beta NdC_2 \rightarrow Nd_2C_3$) also occurs in the Nd–C binary. The temperature of peritectic reaction estimated by Paderno et al. [15] is about 1620 °C.

3. Experiment

Pure neodymium ingots (Alfa Aesar, 99.5 wt% purity) were used in the experiments as the raw materials. Neodymium metal was melted in high purity graphite crucible (30 mm inner diameter and 50 mm height) under argon atmosphere (99.9999%) in an electric resistance furnace with graphite heating elements. Detail description of the experimental facility can be found elsewhere [16].

The samples were heated up to the equilibrium temperatures (1100, 1200, 1350 and 1500 °C, respectively) with heating rate of 18 °C/min (18 K/min) and then holding for a certain time (20–60 min) to reach the equilibrium. Generally speaking, experiments at higher temperatures need shorter time to reach the equilibrium, and vice versa. The samples were cooling down to room temperatures in the inert atmosphere with cooling rate of 22 °C/min (22 K/min). The melting point of neodymium is 1016 °C, the sample are in liquid at the beginning of the equilibrium temperatures. Because the strong affinity of carbon to neodymium, carbon react with neodymium quickly, and the equilibria was reached in a short time usually within the experimental holding time. After carbon saturated, the sample was nearly in solid state, the composition of the C saturated alloy will not be affected by such a quickly cooling process.

The microstructures of all the Nd–C alloy samples were examined with scanning electron microscopy (SU-6600, HITACHI). The carbon contents of the samples were measured with LECO analyzer (CS-444, LECO). Because the Nd–C alloy is very easy to react with moisture in air, the samples were usually protected by the mineral oil. Before the measurements, the oiled samples were washing by acetone for several times to remove the mineral oil.

Table 3
Nominal composition of the Nd–C_{sat} alloy prepared at different temperatures (mass%).

Exp. no.	Temperature (°C)	Holding time (min)	C	Nd
14-1	1500	20	7.46	92.54
14-2	1500	30	7.60	92.40
14-3	1350	30	6.32	93.68
14-4	1200	40	5.13	94.87
14-5	1100	60	4.67	95.33

The carbon contents in the Nd–C_{sat} alloys along with the experimental conditions are tabulated in Table 3. The contents of carbon obtained at 1500 °C for 20 min and 30 min are rather close. It means that carbon dissolves into liquid neodymium melts rather quickly. In order to reach the carbon saturated limit, the holding time for lower temperature samples increase from 30 min to 60 min.

The back-scattering scanning electron microscope (BSEM) images of the carbon saturated Nd–C alloys obtained at different temperatures are shown in Fig. 1. Fig. 1(a)–(d) are the samples gotten at 1100, 1200, 1350, 1500 °C, respectively. Normally, three different phases were observed in the BSEM images: the dark phase, the substrate grey phase and the white phase. The white phase is the pure Nd metal, and the grey phase is Nd metal with carbon saturated. The dark phase is the carbide. From all the samples shown in Fig. 1(a)–(d), only very small amount of white phase can be observed. The predominated phases are the grey Nd–C_{sat} and the dark carbide phases. It is confirmed that the samples were at least close to the carbon saturated limits. From the SEM images, we may also concluded that the carbides with high melting points precipitate primarily. The amounts of carbide in the samples increase with the increase of experimental temperatures.

4. Thermodynamic models

The Gibbs energy of pure element i (i=Nd, C), $G_i^{\varphi}(T) = {}^0G_i^{\varphi}(T) - H_i^{SER}$, in the phase φ is represented as a power series in terms of temperature:

$$ {}^0G_i^{\varphi}(T) = a + b\,T + c\,T\,lnT + d\,T^2 + e\,T^3 + f\,T^{-1} + gT^{-3} + h\,T^{-2} \quad (1) $$

where $a, b, c, \cdots, h$ are the coefficients. H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state. T is the absolute temperature.

The liquid phase is described by the disordered solution phase model. Its excess Gibbs energy is modeled by the Redlich–Kister polynomial. The Gibbs energy of the liquid phase is represented by the following equation:

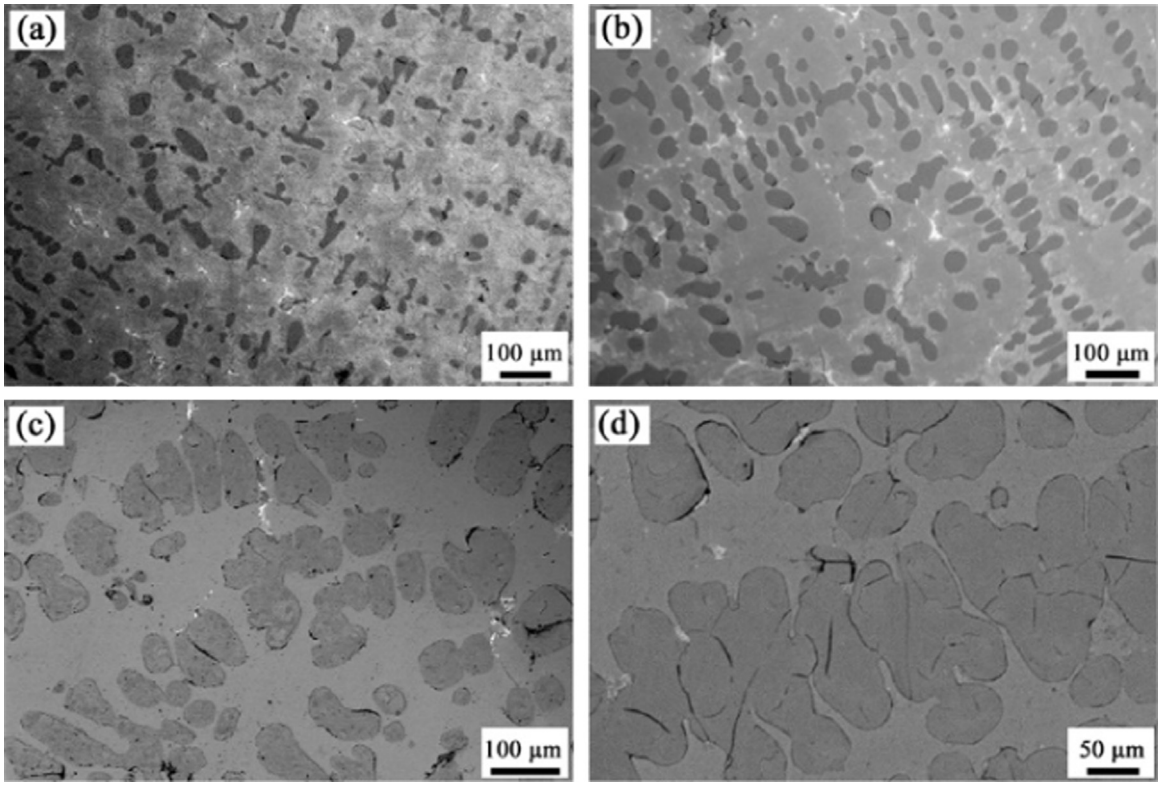

Fig. 1. BSEM images of the Nd–C_{sat} alloys obtained at different temperatures.

$$G^L(T) - H^{SER} = x_C{}^0G_C^L(T) + x_{Nd}{}^0G_{Nd}^L(T) + RT\left[x_C\ln(x_C) + x_{Nd}\ln(x_{Nd})\right] + x_Cx_{Nd}\sum_{k=0}^{n}{}^kL_{C,Nd}^{Liq}(x_C - x_{Nd})^k \quad (2)$$

where H^{SER} is the abbreviation of $x_CH_C^{SER} + x_{Nd}H_{Nd}^{SER}$, R is the gas constant, and ${}^kL_{C,Nd}^{Liq}$ is the kth R–K parameter of liquid phase, which can be expanded as temperature dependence: ${}^kL_{C,Nd}^{Liq} = a_k + b_kT + \cdots$

The solid phases the Nd–C system, Φ (Φ=BCC_A2, DHCP, Nd_2C_3 and βNdC_2) display the range of homogeneity. Since the atomic radius of Nd are much larger than that of C, C is considered in the interstitial position of the lattices. Based on the crystal structure of neodymium, a two-sublattice model, $(Nd)_a(C, Va)_c$, is utilized to model the phases Φ. The Gibbs energy of the phase Φ is expressed by the following equation:

$$G^\Phi - H^{SER} = y_CG_{Nd:C}^\Phi + y_{Va}G_{Nd:Va}^\Phi + cRT\left[y_C\ln(y_C) + y_{Va}\ln(y_{Va})\right] + y_Cy_{Va}\sum_{k=0}^{n}{}^kL_{Nd:C,Va}^\Phi(y_C - y_{Va})^k \quad (3)$$

This quantity can be expressed as $G_{Nd:C}^\Phi - H^{SER} = A + BT + a{}^0G_{Nd}^{ref} + c{}^0G_C^{graphite}$. $G_{Nd:Va}^\Phi$ is the Gibbs energy of the phase Φ in a state when the second sublattice is completely occupied by vacant and can be expressed as $G_{Nd:Va}^\Phi - H^{SER} = {}^0G_{Nd}^{ref}$. $L_{Nd:C,Va}^\Phi$ is the jth parameter of the phase Φ, and can be expressed as $L_{Nd:C,Va}^\Phi = A_j + B_jT$. The interaction parameters A_j and B_j are to be optimized. When the phase Φ stands for αNd, a and c equal to be 1/2, when the phase Φ stands for βNd, a and c equal to be 1/4 and 3/4, respectively; when the phase Φ stands for Nd_2C_3, a and c equal to be 2/5 and 3/5, respectively; when the phase Φ stands for βNdC_2, a and c equal to be 1/3 and 2/3, respectively.

In view of negligible homogeneities for αNdC_2, this phase is treated as stoichiometric ones. The Gibbs energy of each phase per mole-atoms is given by the following expression:

$$G_{Nd_{(1-x)}C_x}^\Phi - H^{SER} = (1-x){}^0G_{Nd:}^{ref} + x{}^0G_C^{graphite} + A + BT + \cdots \quad (4)$$

where A and B are the parameters to be evaluated in the course of optimization. $1 - x$ and x are atomic fractions of Nd and C for αNdC_2.

5. Results and discussion

Based on all available thermodynamic data for the compounds, solid and liquid solutions from the literatures and our experimental works, the optimization of the thermodynamic parameters was carried out by using the Phase diagram module of FactSage software. Using these parameters, most of the thermodynamic property and phase boundary data can be well reproduced. The optimization of the Nd–C phase diagram can be divided in two steps. First, treat the homogeneity phase of Nd_2C_3 and βNdC_2 as stoichiometric phase along with αNdC_2 to fit the evaluated experimental data of the liquid and solid solution phases. All the thermodynamic parameters were optimized to describe the general feature of the phase diagram, for instance, the eutectic points, the melting points of the intermetallic alloys and the liquidus. Secondly, the homogeneity ranges of Nd_2C_3 and βNdC_2 were taken into consideration. Fixing all the thermodynamic parameters gotten in the first step, the solution of carbon in the homogeneity phases were taken into account to present the phase of Nd_2C_3 and βNdC_2. After these two steps, all the thermodynamic parameters of Nd–C system were optimized corresponding to the reliable experimental data, and listed in Table 4.

Fig. 2 shows the calculated phase diagram of the Nd–C system together with the experimental data and all the invariant temperatures are tabulated in Table 5, along with the data obtained from the literatures. It can be seen that all the experimental data

Table 4
Summary of the thermodynamic parameters in the C–Nd systems.

Phase	Model	Parameters
liquid	$(C,Nd)_1$	${}^0L^{Liq}_{Nd,C} = -268285.3 + 22.227T$
		${}^1L^{Liq}_{Nd,C} = 132872.9$
		${}^2L^{Liq}_{Nd,C} = -24515.3$
DHCP (Nd-based)	$(Nd)_1(C,Va)_1$	${}^0G^{DHCP}_{Nd:C} = {}^0G^{DHCP}_{Nd} + {}^0G^{graphite}_{C}$
		${}^0G^{DHCP}_{Nd:Va} = 5000 + {}^0G^{DHCP}_{Nd}$
BCC	$(Nd)_1(C,Va)_3$	${}^0G^{BCC}_{Nd:C} = {}^0G^{BCC}_{Nd} + 3\,{}^0G^{graphite}_{C} + 8314$
		${}^0G^{BCC}_{Nd:Va} = {}^0G^{BCC}_{Nd}$
		${}^0L^{BCC}_{Nd:C,Va} = 5000$
NdC_2	$(Nd)_1(C,Va)_2$	${}^0G^{NdC_2}_{Nd:C} = -42918 - 28.2T + {}^0G^{DHCP}_{Nd} + 2\,{}^0G^{graphite}_{C}$
		${}^0G^{NdC_2}_{Nd:Va} = 60000 + {}^0G^{DHCP}_{Nd}$
Nd_2C_3	$(Nd)_2(C,Va)_3$	${}^0G^{Nd_2C_3}_{Nd:C} = -179255 + 8.75T + 2\,{}^0G^{DHCP}_{Nd} + 3\,{}^0G^{graphite}_{C}$
		${}^0G^{Nd_2C_3}_{Nd:Va} = 100000 + 2\,{}^0G^{DHCP}_{Nd}$

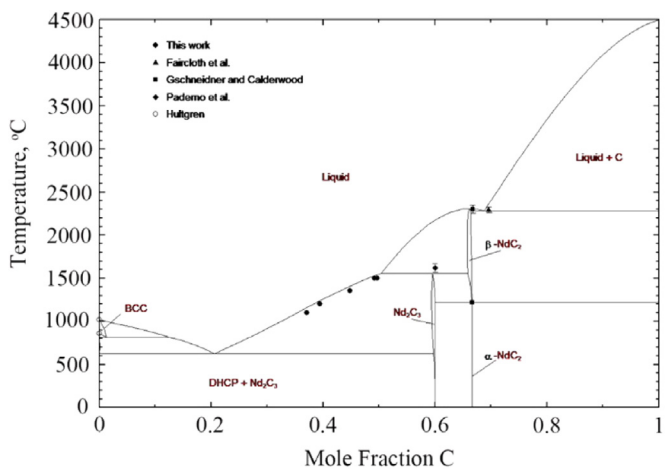

Fig. 2. The calculated C–Nd phase diagram.

Table 5
Calculated invariant equilibria in the Nd–C systems compared with the literature data.

Invariant reactions	T (°C)	Method	Source
$(\beta Nd) \leftrightarrow (\alpha Nd) + L$	810	Calculated	Present work
$L \leftrightarrow (\alpha Nd) + Nd_2C_3$	616	Calculated	Present work
$L + \beta NdC_2 \leftrightarrow Nd_2C_3$	1592	Calculated	Present work
	1620 ± 30	DTA	[15]
$\alpha NdC_2 + C \leftrightarrow \beta NdC_2$	1211	Calculated	Present work
	1150 ± 20	DTA	[1]
$L \leftrightarrow \beta NdC_2 + C$	2263	Calculated	Present work
	2260 ± 30	Estimated	[13]

agree well with the calculated phase diagram. Although there is no available data about the eutectic point of the Nd–Nd_2C_3, by fitting our experimental data with the phase diagram, the eutectic point of Nd–Nd_2C_3 are calculated as at 623 °C with the carbon content of 20.4 at%. In the Nd–C system, the other eutectic point is NdC_2–C with temperature of 2286 °C with carbon content of 68.6 at%. The decomposition of Nd_2C_3 is at the temperature of 1558 °C, and the transition temperature of αNd→βNd is at 1221 °C. From our calculation the C solubility limit in the βNd is about 1.1 at%, that value in αNd is not considered in present work.

The calculated enthalpies of formation in the Nd–C system are compared with the experimental data, as shown in Fig. 3. The

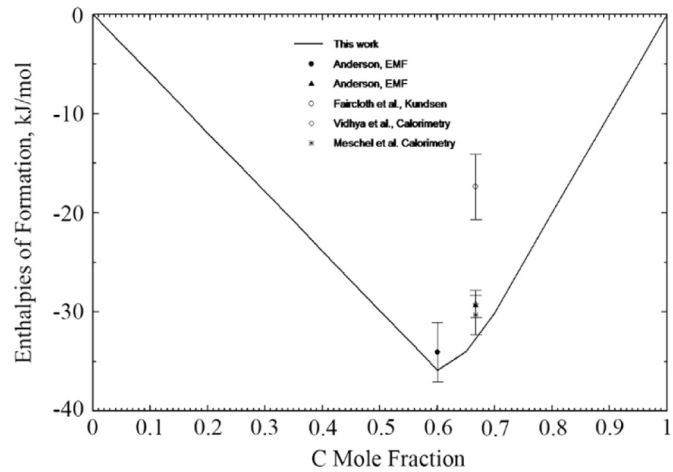

Fig. 3. Enthalpies of formation for the Nd–C system at 298 K.

enthalpy of Nd_2C_3 measured by EMF method [6] agrees well with the calculated value. The enthalpy of NdC_2 measured by synthesis calorimetry [9,10] and EMF [7] are also close to the calculated value.

6. Conclusion

The Nd–C binary system has been assessed based on the experimental investigation and the literature survey. The intermediate compounds of Nd_2C_3 and βNdC_2 (high-temperature form of NdC_2 phase) were considered to be the non-stoichiometric phases whereas the αNdC_2 (low-temperature form of NdC_2 phase) was treated as the stoichiometric phase. Carbon contents of the Nd–Nd_2C_3 range were measured in the present study. Enthalpies of formation of Nd_2C_3 and NdC_2 carbides have been modeled by CALPHAD method. The phase diagram of Nd–C system has been modeled using the FactSage software.

Acknowledgment

The present work was funded by the European Commission (Contract no. 603564) in the REEcover project.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2015.10.001.

References

[1] K.A. Gschneidner Jr., F.W. Calderwood, Bull. Alloy Phase Diagr. 7 (1986) 421–436.
[2] M. El Makrini, D. Guérard, P. Lagrange, A. Hérold, Physica B+C 99 (1980) 481–485.
[3] F.H. Spedding, K. Gschneidner, A.H. Daane, J. Am. Chem. Soc. 80 (1958) 4499–4503.
[4] R. Hultgren, Selected Values of Thermodynamic Properties of Metals and Alloys, Wiley, New York, 1963.
[5] N.H. Krikorian, T.C. Wallace, M.G. Bowman, Phase Relationships of the High-Carbon Portion of Lanthanide–Carbon Systems, Centre National Recherche Scientifique, Paris, 1967.
[6] J.S. Anderson, A.N. Bagshaw, Thermodynamic Data for Some Rare Earth Carbides by Solid-State E.M.F. Techniques, Centre National de la Recherche Scientifique, Paris, 1970.
[7] J.S. Anderson, A.N. Bagshaw, Rev. Chim. Miner. 9 (1972) 115–138.

[8] R.L. Faircloth, R.H. Flowers, F.C.W. Pummery, J. Inorg. Nucl. Phys. 30 (1968) 499–518.
[9] R. Vidhya, M.P. Antony, P.R. Vasudeva Rao, B. Viswanathan, J. Nucl. Mater. 295 (2001) 228–232.
[10] S.V. Meschel, O.J. Kleppa, J. Alloy. Compd. 217 (1995) 235–239.
[11] K.A. Gschneidner Jr., F.W. Calderwood, Bull. Alloy Phase Diagr. 7 (1986) 446–449.
[12] Y. Peng, Y. Du, L. Zhang, C. Sha, S. Liu, F. Zheng, D. Zhao, X. Yuan, L. Chen, CALPHAD 35 (2011) 533–541.
[13] K.A. Gschneidner, F.W. Calderwood, Bull. Alloy Phase Diagr. 7 (1986) 437–438.
[14] K.A. Gschneidner, F.W. Calderwood, Bull. Alloy Phase Diagr. 7 (1986) 558–559.
[15] Yu.B. Paderno, V.L. Yupko, G.N. Makarenko, Izv. Akad. Nauk SSSR Neorg. Mater. 5 (1969) 386–388.
[16] L.K. Jakobsson, M. Tangstad, Metall. Mater. Trans. B 45 (2014) 1644–1655.